



Intuitive interaction for robots among humans

Fact Sheet

Project Information

INTERACT

Grant agreement ID: 101041863

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[10.3030/101041863](https://doi.org/10.3030/101041863) 

EC signature date

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End date

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European Research Council (ERC)

Total cost

€ 1 499 999,00

EU contribution

€ 1 499 999,00

Investment in EU policy priorities

Digital agenda  Clean air 

Artificial Intelligence  Climate action 

Biodiversity 

Coordinated by

TECHNISCHE UNIVERSITEIT
DELFT

Netherlands

Project description

Laying the foundations for intuitive multi-robot interaction

Mobile robots can move around in the environment and are not bound to one physical location. While much progress has been made in the field, these robots are still not capable of seamless interaction in human-populated environments. The EU-funded INTERACT project aims to address this challenge. Taking a holistic view on the interaction of mobile robots and humans, the project aims to provide mobile robots with the ability to reason about their coordination with other agents, evaluate associated risks, continuously reassess their environment and take appropriate actions to achieve their tasks. The project's results will have significant impacts on automated processes in industrial and urban settings.

Objective

Thanks to developments in computing, sensing and communication technology, great progress has been made in the control of robots affixed to the factory floor, mobile robots in controlled environments, and self-driving cars in medium-complexity environments. However, mobile robots are not yet capable of seamless interaction in human populated environments, primarily due to the high complexity of modeling and reasoning over the effect that robot actions have on other robots and humans, and the inherent uncertainty. Humans rely on intuition, namely the ability to understand interaction instinctively, without the need for conscious reasoning. However, for control systems it is notoriously difficult to model intuition, and to do so with safety guarantees with respect to physical constraints. My goal is to provide mobile robots with the ability to reason about their coordination with other agents and the associated risks, take appropriate actions and continuously reassess.

I will reach this goal by developing an innovative approach across the boundaries of Motion Planning, Multi-Robot Task Assignment and Machine Learning. In INTERACT I propose a holistic view on the interaction of mobile robots and humans, where I will consider multiple spatio-temporal granularities ranging from individual interactions to the interaction of a robot fleet with humans in a city, and from short term (local) to long term (global) effects of the interaction. Robots will use past experience to learn local and global intuition models of their interaction with the environment. These intuition models will be integrated in novel uncertainty-aware optimization methods to compute safe interaction-aware trajectories, task

assignments and routes for mobile robots.

INTERACT will lay the foundation for intuitive multi-robot interaction, make it possible for teams of mobile robots to safely interact in human-centric environments and enable a new level of automation in factories and cities.

Fields of science (EuroSciVoc) ⓘ

[engineering and technology](#) > [mechanical engineering](#) > [vehicle engineering](#) > [automotive engineering](#) > **[autonomous vehicles](#)**

[engineering and technology](#) > [civil engineering](#) > [urban engineering](#) > **[smart cities](#)**

[engineering and technology](#) > [electrical engineering, electronic engineering, information engineering](#) > [electronic engineering](#) > **[control systems](#)**

[engineering and technology](#) > [electrical engineering, electronic engineering, information engineering](#) > [electronic engineering](#) > [robotics](#) > **[autonomous robots](#)**



Keywords ⓘ

[Multi-robot systems](#)

[Motion planning](#)

[Dynamic vehicle routing](#)

[Intelligent transportation](#)

Programme(s) ⓘ

HORIZON.1.1 - European Research Council (ERC)

MAIN PROGRAMME

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Topic(s)

[ERC-2021-STG - ERC STARTING GRANTS](#) 

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Funding Scheme

HORIZON-ERC - HORIZON ERC Grants

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Call for proposal

[ERC-2021-STG](#) 

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Host institution



TECHNISCHE UNIVERSITEIT DELFT

Net EU contribution 

€ 1 499 999,00

Total cost 

€ 1 499 999,00

Address

STEVINWEG 1
2628 CN DELFT

Netherlands



Activity type

Higher or Secondary Education Establishments

Links

[Contact the organisation](#)

[Website](#)

[Participation in EU R&I programmes](#)

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Beneficiaries (1)



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€ 1 499 999,00

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